

Matching the Statistic to the Hypothesis: Geometric-Null Binomial Proximity Testing for Harmonic Longitude Structure in Monument Distributions

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Abstract

Claims that ancient or sacred monuments align with longitude grids raise two separable questions: whether a spatial pattern is statistically detectable, and how such a pattern should be interpreted archaeologically. Following [Silva \(2020\)](#), this paper treats the first as a problem of statistical description and the second as a separate historical, cultural, and geographic task; the methodological focus is matching the statistic to the hypothesis. The Rayleigh statistic, a standard circular-statistical diagnostic in archaeoastronomy, measures mean phase concentration at period T . A thresholded repeated-line hypothesis, however, predicts proximity to any harmonic longitude within a specified tolerance. These are distinct quantities. Rayleigh R remains useful for detecting circular phase concentration, but it can dissociate from harmonic proximity: a concentrated off-harmonic cluster can produce large R , while many near-harmonic sites mixed with off-harmonic mass can yield proximity enrichment with weak Rayleigh concentration.

The matched statistic for “sites cluster near harmonics of period T ” is a one-sided binomial count of sites within distance d of any harmonic, benchmarked against the geometric null $p_0 = 2d/T$. The framework is demonstrated under a fixed operational design: fixed period, fixed corridor phase (the anchor longitude λ_0 defining the harmonic grid), fixed thresholds, and fixed monument corpora, including a UNESCO morphology-term keyword corpus defined by words denoting dome-like or hemispherical architecture. In the UNESCO dome/stupa corpus ($N = 90$), a period sweep shows the Rayleigh–proximity dissociation directly: Rayleigh significance appears at 2.4° , 3.5° , and 3.6° without corresponding proximity enrichment, while the enriched 3.0° period is Rayleigh-non-significant. Applied to three operational hemispherical-monument corpora at the fixed 3° period and $\sim 35.25^\circ\text{E}$ Levantine corridor phase, the geometric-null binomial tests yield a consistent corpus-level harmonic-proximity pattern in the worked example, summarized by Fisher’s method ($p < 0.001$ at Tier A). The principal contribution is a matched statistical framework for thresholded harmonic-proximity hypotheses. It is demonstrated through an open-data worked example that illustrates the Rayleigh–proximity dissociation and the complementary roles of the two statistics.

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1 Introduction

Do monuments cluster near specific longitude intervals? The question recurs in the archaeoastronomical and archaeogeographic literature, usually motivated by observational coincidences involving sacred sites, trade corridors, or cosmographic traditions. It has also been criticized on grounds of investigator-assembled site lists, insufficient correction for multiple comparisons, and lack of independent replication (Williamson & Belamy, 1983; Freeman, 1976; Kendall, 1974). These concerns motivate a stricter separation between statistical detection and archaeological explanation. Alongside these critiques, this paper raises a complementary issue of test-hypothesis matching: the standard circular-statistical tool used in this literature measures a different quantity than the one a repeated-line grid hypothesis predicts.

The tool in question is the Rayleigh statistic R . R measures the degree to which a set of phases concentrates toward a single preferred direction on the unit circle — it is a sensitive test for unimodal directional concentration. Prior studies have applied Rayleigh tests to longitude distributions and interpreted a significant R at some period T as evidence that sites are placed near the harmonics of T (Thom, 1967; Magli, 2013, 2016).

This is a useful diagnostic but only a partial answer to the repeated-line question. A grid-placement hypothesis predicts that sites fall near longitudes $\lambda_0, \lambda_0 + T, \lambda_0 + 2T, \dots$. Under modulo- T phase mapping these harmonics collapse to a common phase, so an ideal grid-aligned corpus would produce strong Rayleigh concentration. In real geographically clustered corpora, however, Rayleigh concentration and thresholded harmonic proximity can dissociate: a dominant off-harmonic cluster can produce significant Rayleigh concentration, while a corpus with many near-harmonic sites plus substantial off-harmonic mass can produce binomial enrichment without significant Rayleigh R . A non-significant Rayleigh result therefore does not rule out harmonic proximity, and a significant Rayleigh result does not by itself establish proximity to repeated longitude lines. The two statistics are complementary; they answer related but distinct questions.

The directly matched test for “sites cluster near harmonics” is straightforward. Under a uniform longitude model, the expected fraction of longitudes falling within $\pm d^\circ$ of any harmonic of period T is $p_0 = 2d/T$. This is the *geometric null*. In clustered corpora it remains a useful geometric benchmark, but the resulting p -values are interpreted here as corpus-level proximity enrichment rather than counts of independent placement decisions.

This paper develops that matched test and uses the monument data as a worked methodological example. Under a fixed period, fixed corridor phase, fixed proximity thresholds, and fixed operational corpora, the geometric-null binomial test estimates whether more sites fall near harmonic longitudes than expected from the geometric coverage of the threshold bands. Rayleigh statistics are retained as complementary circular diagnostics, and the period sweep is used to show when phase concentration and thresholded harmonic proximity agree or dissociate. The empirical application illustrates corpus-level proximity enrichment at the fixed 3° period and $\sim 35.25^\circ\text{E}$ Levantine corridor phase. Following Silva (2020), this is treated throughout as statistical description

of spatial structure in an open-data worked example; historical, cultural, or geographic explanation is a separate interpretive task requiring independent evidence.

2 Rayleigh Concentration and Harmonic Proximity: Complementary but Distinct Statistics

2.1 Why the Two Statistics Can Dissociate

Let $\lambda_1, \dots, \lambda_N$ be site longitudes and T a candidate grid period with anchor λ_0 . Define phases by $\theta_i = 2\pi((\lambda_i - \lambda_0) \bmod T)/T$, and the Rayleigh statistic by

$$R = \left| \frac{1}{N} \sum_{i=1}^N e^{i\theta_i} \right|.$$

A key property of this transformation is that sites near *any* harmonic of T — near λ_0 , near $\lambda_0 + T$, near $\lambda_0 + 2T$, and so on — all map to the same phase $\theta \approx 0$ under the modulo- T reduction, because $(\lambda_0 + kT - \lambda_0) \bmod T = 0$ for any integer k . Under ideal conditions (all sites falling exactly on a harmonic), R would equal 1. The modulo- T phase map therefore does not, by itself, produce multiple phase classes: harmonics collapse to one class, not many.

The Rayleigh test and the binomial proximity test can nevertheless dissociate because they aggregate cross-site information differently.

What Rayleigh measures. R is the length of the mean phasor: it is sensitive to whether a *single direction* on the unit circle predominates after averaging over the full corpus. In a geographically clustered corpus, a single dominant geographic cluster can produce large R at period T simply because that cluster’s longitude maps to a concentrated phase at that particular T — regardless of whether the cluster is harmonically proximate. This is a generic property of clustered data on the unit circle; it can produce significant R at periods that carry no harmonic-proximity interpretation.

What the binomial measures. The binomial proximity test counts the number of sites whose deviation from the nearest harmonic falls within a threshold d : $\#\{i : \delta_i \leq d\}$. Let $r_i = ((\lambda_i - \lambda_0) \bmod T)$ and $\delta_i = \min(r_i, T - r_i)$. This is independent of the mean phase direction. Sites far from harmonics (large δ_i) contribute zero to the binomial count regardless of whether they push R up or down. Conversely, a corpus can show significant binomial enrichment even when many off-harmonic sites keep the mean phasor small.

Complementarity. The two tests ask complementary questions. Rayleigh asks: does one phase direction dominate? The binomial asks: are more sites within a fixed reporting threshold of a harmonic than the uniform null predicts? In a real corpus these can agree or disagree, depending on the balance between harmonically proximate sites, off-harmonic sites, and the geographic clustering structure of the corpus. A non-significant Rayleigh result does not rule out harmonic proximity; a significant Rayleigh result does not establish harmonic proximity unless the phase concentration is verified to coincide with a harmonic. Relying on Rayleigh alone to evaluate a repeated-line hypothesis can therefore produce false positives at cluster-dominated periods and miss genuine proximity enrichment when near-harmonic sites coexist with substantial off-harmonic mass.

2.2 Empirical Demonstration: Period Sweep Dissociation

Table 1 illustrates the dissociation concretely using the UNESCO keyword-matched dome/stupa corpus ($N = 90$). Four candidate periods are shown, selected to represent the full range of outcomes: three Rayleigh-significant periods without proximity enrichment, and one period with proximity enrichment but no Rayleigh significance. Both tests use the Tier-A threshold ($\pm 0.10 \times T$, geometric null $p_0 = 20\%$); finer-threshold tiers (A+, A++) are reported in the primary results.

Table 1: Rayleigh concentration vs. Tier-A binomial proximity enrichment at four candidate periods, UNESCO dome corpus ($N = 90$, corridor phase $\sim 35.25^\circ\text{E}$). Tier-A threshold = $0.10 \times T$; geometric null $p_0 = 20\%$. The two statistics dissociate across periods: three periods show Rayleigh significance without proximity enrichment (2.4° , 3.5° , 3.6°), while the period with the strongest proximity enrichment (3.0°) is Rayleigh-non-significant. This illustrates that the two tests measure different aspects of the phase distribution; neither subsumes the other.

Period	Rayleigh p	Sig	Tier-A hits	Expected	Binom. p	Sig
2.4°	0.0083	**	21	18.0	0.2503	ns
3.0°	0.1796	ns	29	18.0	0.0043	**
3.5°	0.0267	*	13	18.0	0.9312	ns
3.6°	0.0407	*	20	18.0	0.3383	ns

The dissociation is clear. Rayleigh is significant at 2.4° , 3.5° , and 3.6° , and non-significant at 3.0° . The binomial proximity test is significant only at 3.0° . A researcher relying on Rayleigh alone to evaluate the repeated-line hypothesis would flag periods that carry no proximity enrichment, and would overlook the period with the clearest proximity structure. Using the two tests together avoids this.

The pattern is consistent with the mechanism described above. At 2.4° , 3.5° , and 3.6° , one geographic cluster dominates the phase distribution, producing concentrated R without harmonic proximity enrichment. At 3.0° , many sites fall near the harmonic class, but substantial off-harmonic mass keeps the mean phasor small; the binomial test correctly registers the thresholded per-site proximity counts. Rayleigh and the binomial measure genuinely different aspects of the data; neither is wrong, but only the binomial directly answers the question “do sites fall near harmonics?”

A broader candidate-period survey of the dome/stupa corpus (1.5° to 6.0° in 0.1° steps, 46 periods; four representative periods shown in Table 1) finds the dominant binomial proximity peak at 3° ($p = 0.0043$); selected nearby non-harmonic spacings (1.8° , 2.1° , 2.7° , 3.3°) are non-significant in the same sweep. This motivates 3° as the primary period for the cross-corpus analysis in Section 5.

A Monte Carlo spacing test (100,000 permutations, dome-label shuffle) evaluates whether the number of significant proximity peaks across 18 candidate spacings spanning 0.5° to 15° would be surprising under geographic concentration alone. The 18 spacings are: 0.5° to 6.0° in 0.5° steps, plus 7° , 8° , 9° , 10° , 12° , and 15° —chosen to span the plausible range of geographic grid scales without reference to their individual test outcomes; no formal pre-registration was performed. At $\alpha = 0.05$ the naive binomial expectation is $0.05 \times 18 = 0.9$ significant spacings; the Monte Carlo simulation yields a mean of 1.17 (reflecting dependence across spacings). The observed count of 4 significant spacings is marginal ($p \approx 0.0979$). Of the four significant spacings, three (6° , 8° , 12°) are multiples

of 3° : a node-overlap analysis shows that all of their hits are accounted for by the 3° harmonic structure, so their off-node components are non-significant. The 3° spacing is the only one whose significance is not explained by overlap with another spacing’s nodes.

3 The Geometric-Null Binomial Framework

3.1 The Geometric Null

For a grid of period T anchored at λ_0 , define the anchor-relative remainder and nearest-harmonic deviation as

$$r_i = ((\lambda_i - \lambda_0) \bmod T), \quad \delta_i = \min(r_i, T - r_i).$$

A site falls within tier threshold d if $\delta_i \leq d$. Under a uniform longitude model, the expected fraction of longitudes satisfying that inequality is $p_0 = 2d/T$. In clustered corpora this remains a useful geometric benchmark, but the resulting p -values are interpreted here as corpus-level proximity enrichment rather than counts of independent placement decisions.

Geographic clusters, serial nominations, and shared cultural traditions are present in all three corpora. Retaining them preserves the operational corpus definition and avoids imposing an external, theory-laden decision about the relative importance of major monument nodes versus smaller dependent sites. If a major stupa center generated many associated monuments, that multiplicity is part of the observed historical-geographic pattern being measured. The binomial test therefore evaluates proximity enrichment in the corpus as documented, as a corpus-level measure rather than a count of independent intentional placement decisions.

The two null models reported in Section 5 (Null A: within-dome bootstrap; Null B: same-footprint UNESCO background, the discriminating comparison) provide complementary assessments of the observed UNESCO proximity enrichment; they are discussed in Section 6.

3.2 Tier Classification

We define three proximity thresholds as round fractions of the $T/2$ half-period (1.5°), calibrated to bracket the empirical natural proximity scales of the corpora under study: the OWTRAD Silk Road network (2.6%), the UNESCO dome subset (3.6%), the full UNESCO Cultural/Mixed corpus (10.2%), and the Wikidata stupa corpus (20.2%).¹ Each threshold corresponds to a geometric null rate $p_0 = 2d/T$:

Tier	Threshold	Geometric null p_0
A++	$\delta \leq 0.06^\circ$ ($\lesssim 6.66$ km equatorial)	4%
A+	$\delta \leq 0.15^\circ$ ($\lesssim 16.6$ km)	10%
A	$\delta \leq 0.3^\circ$ ($\lesssim 33$ km)	20%

Tier labels are descriptive shorthand; the primary test is always the binomial.

¹These values are the peak-enrichment null rates from the multiscale window sweep (`analysis/unesco/multiscale_combined_p.py`); a standalone reproduction is provided in `supplementary/audit/tier_calibration_audit.py`.

3.3 Cross-Corpus Consistency via Fisher Combination

For each corpus c at tier d , a one-sided binomial test yields p_c . Fisher’s combination statistic $\chi^2 = -2 \sum_c \ln p_c$, referred to $\chi^2(2C)$ for C corpora, summarizes whether the directional excess is consistent across all corpora simultaneously. Fisher’s method formally assumes independent p -values. The three corpora were assembled by distinct processes and their geographic clusters are substantially non-overlapping, but they share historical and geographic structure (Eurasian corridor, stupa tradition, public data sources). The combined p -value is therefore used as a cross-corpus consistency summary, noting that full statistical independence of the three corpora cannot be assumed. Rayleigh statistics are retained as complementary diagnostics for comparability with prior literature.

4 Data

UNESCO dome/stupa subset ($N = 90$). The UNESCO World Heritage XML dataset (UNESCO World Heritage Centre, 2024) contains 1011 Cultural and Mixed inscribed sites. Seven morphological keywords (stupa/stupas, tholos, dome/domed/domes, spherical) were matched against site names, XML descriptions, and OUV text, yielding $N = 90$ sites. All statistical tests use this keyword-matched population.

Wikidata Q180987 stupa corpus ($N = 229$). All *instance-of: stupa* (Q180987) items in the Wikidata knowledge base, extracted via SPARQL query on the *instance-of: stupa* property. The corpus is geographically dominated by the Java/Sumatra node ($\sim 105\text{--}115^\circ\text{E}$), reflecting the documented historical density of stupa construction around Borobudur (Miksic, 1990).

OSM stupa layer ($N = 259$). OpenStreetMap features with tags `building=stupa`, `historic=stupa`, or `ruins=stupa`, assembled by crowd-sourced tagging. Coverage correlates with mapping effort. Uneven contributor density is a coverage limitation; the OSM corpus is therefore treated as supplementary data alongside the primary corpora.

All three corpora were tested against the same fixed candidate frame: period 3° and the $\sim 35.25^\circ\text{E}$ Levantine corridor phase. Because open datasets are publicly available, the frame is treated uniformly as fixed for the present analysis across all three corpora. The empirical application therefore functions as a frame-specified spatial test.

Candidate frame. The candidate reference phase is the $\sim 35.25^\circ\text{E}$ Levantine corridor phase. For reference, this longitude is labeled by Mount Gerizim at 35.269°E , a non-corpus topographic point in the central Levant. Gerizim is not included in the monument corpora. It functions only as a convenient geographic label for the corridor-level phase near Jerusalem and the central Levant. The statistical result is band-invariant: longitudes within the same $\sim 25^\circ\text{E}$ band produce the same proximity enrichment, and the data cannot distinguish Gerizim from Jerusalem (35.232°E) or from other co-grid longitudes within the same proximity tier.

5 Results

5.1 Primary Result: Cross-Corpus Geometric-Null Test

Table 2 reports the primary statistical results: one-sided binomial tests and Fisher combinations across all three corpora.

Table 2: Cross-corpus geometric-null harmonic-proximity enrichment at the fixed 3° period and $\sim 35.25^\circ\text{E}$ Levantine corridor phase. For each corpus and tier, the one-sided binomial test compares the observed number of sites within distance d of any harmonic with the expected count under the geometric benchmark $p_0 = 2d/T$. The resulting p -values are read as corpus-level proximity-enrichment measures under this benchmark, not as counts of independent placement decisions. Fisher’s statistic $\chi^2 = -2\sum_c \ln p_c$, referred to $\chi^2(6)$ for the three corpora, is reported as a cross-corpus consistency summary (see Section 3 for the independence caveat).

Tier	Corpus	N	Hits	Exp	Fold	p_{binom}	Sig
A++ ($p_0=4\%$)	UNESCO domes	90	11	3.6	$3.06\times$	0.0009	***
	Wikidata stupas	229	11	9.2	$1.2\times$	0.311	ns
	OSM stupas	259	12	10.4	$1.16\times$	0.3433	ns
	<i>Fisher $\chi^2(6) = 18.402$, nominal combined $p = 0.0053$</i>						
A+ ($p_0=10\%$)	UNESCO domes	90	18	9.0	$2.0\times$	0.0032	**
	Wikidata stupas	229	33	22.9	$1.44\times$	0.0212	*
	OSM stupas	259	33	25.9	$1.27\times$	0.089	
	<i>Fisher $\chi^2(6) = 24.017$, nominal combined $p = 0.00052$</i>						
A ($p_0=20\%$)	UNESCO domes	90	29	18.0	$1.61\times$	0.0043	**
	Wikidata stupas	229	70	45.8	$1.53\times$	0.0001	***
	OSM stupas	259	69	51.8	$1.33\times$	0.0059	**
	<i>Fisher $\chi^2(6) = 39.661$, nominal combined $p < 0.001$</i>						

In the worked example, all three corpora exceed the geometric null at every tier despite substantially different geographic footprints and assembly processes. Fisher combinations are reported as cross-corpus consistency summaries (see Section 3 for the independence caveat). The UNESCO A++ result shows the largest individual fold-enrichment ($3.06\times$), reflecting the distributed character of that corpus; Wikidata and OSM A++ results are individually non-significant, reflecting cluster-level rather than individual-site precision. The Tier A Fisher combination ($p < 0.001$) reflects the broadest cross-corpus agreement, with all three corpora showing directional excess at this tier. All p -values are frame-conditional proximity-enrichment measures for the fixed 3° period and $\sim 35.25^\circ\text{E}$ Levantine corridor phase; the anchor-sweep correction and conditionality discussion are in Section 6.

5.2 Supplementary Rayleigh Phase Concentration

Table 3 reports Rayleigh statistics at $T = 3^\circ$ for comparability with prior literature. The full UNESCO corpus shows no Rayleigh phase concentration ($R = 0.0224$, perm- $p = 0.6008$, ns) at $T = 3^\circ$, consistent with the proximity pattern not being a generic circular-statistical property of the full heritage list. Watson’s U^2 and Kuiper tests are used here as corroborating two-sample diagnostics (not independent tests): both show that the dome/stupa phase distribution differs detectably from the full UNESCO corpus ($p = 0.0036$, $p = 0.0013$ respectively), consistent with subset-specific enrichment relative to the full UNESCO background and illustrating how two-sample circular tests can serve as a complementary subset-specificity check.

Table 3: Supplementary Rayleigh phase concentration at $T = 3^\circ$, retained for comparability with prior literature. The binomial proximity result (Table 2) is the matched test; Rayleigh statistics are reported here as supplementary diagnostics. All results are conditional on the fixed corridor-phase frame.

Corpus	N	R	perm- p	Sig
UNESCO full corpus	1011	0.0224	0.6008	ns
UNESCO dome/stupa	90	0.1365	0.1883	ns
Wikidata Q180987	229	0.1669	0.0016	**
OSM stupa layer	259	0.1073	0.0355	*
<i>Fisher (Wikidata + OSM): $\chi^2(4) = 19.551, p = 0.0006$</i>				

5.3 Negative Controls and Scope

The OWTRAD Silk Road route network ($N = 1674$ nodes) does not independently recover the monument phase in the Normalized Phasor Consensus (NPC) test ($K =$ modulus of dot product of two normalized mean phasors, where $K = 1$ indicates identical phase direction and $K = 0$ orthogonal phases; formally defined in Section 5.4; $K = 0.7429$, perm- $p = 0.46532$, ns). Using the 1946 edge midpoints as the proximity sample — one value per route segment, non-deduplicated, the most conservative representation of the network — the binomial test is non-significant at both tiers (A++: $p = 0.0908$, ns; A+: $p = 0.0679$, ns). The route network therefore shows no enrichment under either the directional or the per-segment proximity test when each segment is counted once.

A distinct question is whether *high-centrality* route hubs — cities where many independent routes converge — fall preferentially near harmonics. Degree-weighting each city by the number of edges it connects ($N = 3892$ node-in-edge occurrences) is one proxy for hub centrality: a city connecting six routes contributes six counts, reflecting its structural prominence in the network. Under this weighting the binomial enrichment is significant in the worked example (A++: $p < 0.0001$, $z = 4.53$; A+: $p = 0.0002$), illustrating that degree-weighting shifts the test output: the most network-central cities fall near the 3° harmonic grid more often than the uniform null predicts under this frame-conditional weighting.

The UNESCO Americas sub-corpus ($N = 145$) shows A+ rate of 9.7%, below the 10% geometric null, indicating no proximity enrichment at the $\sim 35.25^\circ\text{E}$ Levantine corridor phase in the Western Hemisphere.

A type-matched geographic control uses the North American mound corpus ($N = 236$): National Historic Landmark (NHL)- and National Register of Historic Places (NRHP)-registered earthen burial and platform mounds assembled from the NPS National Register database with coordinates resolved via Wikidata (NPS, 2024; Wikimedia Foundation, 2024). These are hemispherical structures morphologically comparable to the Eurasian corpora but located entirely in the Western Hemisphere, outside any connection to the $\sim 35.25^\circ\text{E}$ Levantine corridor phase. At the 3° period and $\sim 35.25^\circ\text{E}$ Levantine corridor phase the mounds corpus shows no proximity enrichment at any tier: A++: 1/236 (0.4%, null 4%, $p = 1.00$, ns); A+: 11/236 (4.7%, null 10%, $p = 1.00$, ns); A: 45/236 (19.1%, null 20%, $p = 0.66$, ns). All three tiers are at or below the geometric null. This supports the methodological illustration that the Eurasian pattern is geographically specific to the Eurasian corridor rather than a generic outcome of applying the same

harmonic-proximity test to hemispherical monument corpora.

Null A (within-dome bootstrap, 100,000 trials) resamples from dome/stupa sites' own longitude distribution; the observed A++ count is not significant against this null ($p = 0.5493$, ns); see Section 6 for interpretation.

Null B (same-footprint restricted draw, 100,000 trials) asks a narrower subset-specificity question: among all UNESCO cultural sites whose longitude falls within $\pm 5^\circ$ of any dome site longitude, does the dome/stupa subset show greater harmonic proximity than the broader background? The broader same-footprint UNESCO background expects only ~ 5.1124 A++ hits versus the observed 11 ($p = 0.0129$ at the $\pm 5^\circ$ window, surviving Bonferroni correction for $k = 3$ windows tested, $\alpha_{\text{adj}} = 0.0167$), illustrating subset-specific enrichment within that geography relative to the broader background. Nulls A and B are discussed further in Section 6.

5.4 Anchor Concordance

All three corpora recover the same $\sim 35.25^\circ\text{E}$ Levantine corridor phase. The Normalized Phasor Consensus statistic K is defined as $K = |\bar{u}_A \cdot \bar{u}_B|$, where $\bar{u}_c = (\sum_i e^{i\theta_i^{(c)}}) / |\sum_i e^{i\theta_i^{(c)}}|$ is the normalized mean phasor of corpus c at the candidate period, and $|\cdot|$ denotes the modulus of the dot product of two unit complex numbers. K compares phase directions independently of tier counts: $K = 1$ when the two corpora point to the same phase, $K = 0$ when they are orthogonal. In the worked example, $K \geq 0.9675$ for all pairings of the three monument corpora (Table 4). The three-corpus combination reaches perm- $p = 0.02437$ with consensus anchor at 0.029° from 35°E . The primary non-nested two-corpus pair (Dome + Wikidata) is individually non-significant, reflecting shared Eurasian geography; significance emerges when the independently assembled OSM corpus is added as a third, geographically distinct data source. Negative controls show K drops sharply when the monument corpora are replaced with geographic or route-network controls.

Table 4: Selected NPC anchor-concordance configurations at $T = 3^\circ$ (100,000 permutation draws). High K with non-significant two-corpus perm- p reflects shared Eurasian geography. The ‘‘Conservative’’ row pairs the broadest possible corpus (full UNESCO Cultural/Mixed) with the Wikidata stupa corpus, maximizing denominator size at the cost of corpus specificity. Negative controls (Rows 4–5) indicate K is not a generic Eurasian property.

Role	Corpora	K	perm- p	Note
Non-nested pair	Dome + Wikidata Stupa	0.9675	0.16287	Primary non-nested
Conservative	UNESCO full + Stupa	0.982	0.12243	Broadest corpus pair
Three-corpus	Dome + Stupa + OSM	0.972	0.02437	All 3 monument corpora
Neg. control	UNESCO + Americas + Stupa	0.6723	0.29819	Geographic control
Contrast corpus	UNESCO + OWTRAD	0.7429	0.46532	Route network
Type control	N. American mounds ($N = 236$)	—	—	No proximity enrichment

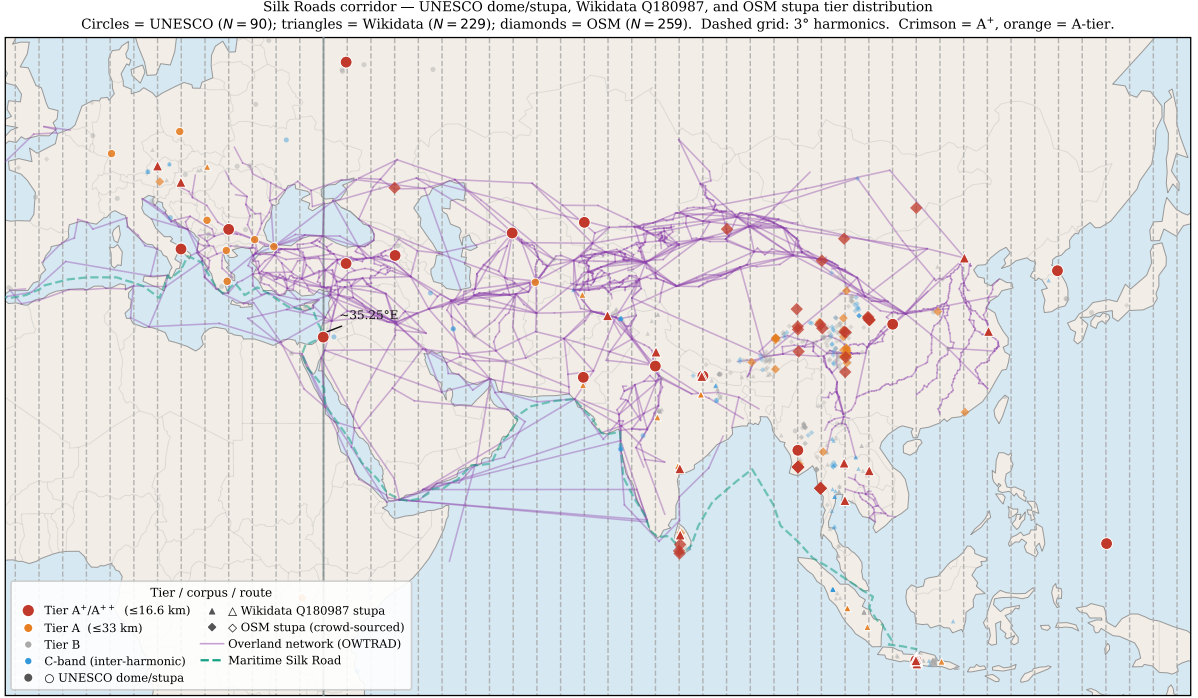


Figure 1: Silk Roads corridor showing all three monument corpora mapped against the 3° harmonic grid (dashed lines) anchored to the $\sim 35.25^\circ\text{E}$ Levantine corridor phase. Circles = UNESCO dome/stupa ($N = 90$, keyword-matched); triangles = Wikidata Q180987 stupa ($N = 229$); diamonds = OSM stupa ($N = 259$, crowd-sourced). Color indicates harmonic proximity tier: crimson = Tier A^+ or better (≤ 16.6 km), orange = Tier A (≤ 33 km), gray/blue = B/C-band. The dark meridian marks the fixed $\sim 35.25^\circ\text{E}$ corridor phase. All results are conditional on the fixed corridor-phase frame.

6 Discussion

6.1 The Methods Contribution

The central methodological result is the separation between circular phase concentration and thresholded harmonic proximity. Rayleigh R describes whether the full phase distribution has a dominant mean direction at period T ; the geometric-null binomial describes whether a corpus contains more sites within a stated operational distance of any harmonic than expected from the fraction of longitude covered by those bands. Because the two statistics weight off-harmonic observations differently, they provide complementary information and can dissociate within the same corpus. This makes Rayleigh a useful diagnostic for phase structure and the geometric-null binomial the matched statistic for thresholded proximity-to-harmonics claims. In Silva’s terms, the tests detect and describe structured features in the spatial data; archaeological, historical, or geographic explanations for those features remain interpretive questions to be evaluated with independent evidence.

The analysis is a descriptive spatial test of harmonic proximity. Geographic clustering is part of the documented corpus pattern, so site-level binomial p -values describe corpus-level proximity enrichment rather than independent cultural decisions. The geometric null $p_0 = 2d/T$ gives a transparent expected hit fraction under uniform longitude, and Fisher’s method is used as a cross-corpus consistency summary. This framework is applicable to

any spatial periodicity hypothesis of this form regardless of geographic or cultural context.

6.2 Interpretation of the Empirical Illustration

The three-corpus worked example finds consistent directional excess over the geometric null at the fixed 3° period and $\sim 35.25^\circ\text{E}$ Levantine corridor phase (Fisher $p < 0.001$ at Tier A). All results are frame-conditional proximity-enrichment measures (see Limitations). The UNESCO corpus is the most geographically distributed result, with A-tier sites spanning the full Eurasian corridor.

Null A and Null B address the two tractable aspects of what the observed UNESCO proximity pattern means. Null A (within-dome bootstrap) is reported for completeness only. Because it resamples from the observed dome-site positions, its expected hit rate is defined by the same geography that carries the observed proximity enrichment — it is not a substantive alternative null model. Its sole informational function is to show that the enrichment is localized to dome-site geography, not an artifact of the comparison procedure ($p = 0.5493$, ns).

Null B provides the discriminating comparison: among all UNESCO cultural sites occupying the same longitude footprint, is the dome/stupa subset more harmonically proximate than the broader background? In the worked example, the dome/stupa subset does show greater proximity than the broader background ($p = 0.0129$, surviving Bonferroni correction), illustrating how Null B functions as the subset-specificity check. The cross-corpus phase concordance (NPC $K \geq 0.9675$ across all pairings of three independently assembled monument datasets with different geographic centers of mass) is consistent with the phase not being an artifact of any single corpus’s geographic distribution: UNESCO dome sites, Wikidata Q180987 stupas, and OSM stupa tags converge on the same $\sim 35.25^\circ\text{E}$ phase independently of one another.

Corpus overlap and curation. The three corpora are assembled by distinct processes: the UNESCO dome/stupa subset derives from keyword matching against the WHC XML dataset; Wikidata Q180987 was extracted via SPARQL query on the *instance-of: stupa* property; OSM was assembled by crowd-sourced tagging of `building=stupa`, `historic=stupa`, and `ruins=stupa` features via Overpass API. A best-effort empirical check using GPS proximity (threshold < 1 km) and name-string matching finds 4 near-coincident site pairs between UNESCO and Wikidata, 0 between UNESCO and OSM, and 1 between Wikidata and OSM (see supplementary overlap audit). Because all three datasets are publicly available, the candidate frame is handled as fixed across the empirical demonstration. The Fisher combination is used as a consistency check across differently assembled datasets. The geographic clusters of the three corpora are substantially non-overlapping at the region level: UNESCO dome sites span the full Eurasian corridor from the Levant to Java; Wikidata is dominated by the Java/Sumatra node; OSM coverage reflects crowd-sourced mapping density across South and Southeast Asia. No single shared geographic cluster explains enrichment in all three corpora simultaneously.

The Wikidata result is substantially carried by the Java/Borobudur node, which is historically the dominant node of the stupa tradition (Miksic, 1990): the cluster’s position on a 3° harmonic is the proximity pattern the geometric-null test is designed to detect. The OSM result, assembled by a process with no relationship to UNESCO or Wikidata curation, shows the same directional excess ($p = 0.0059$ at Tier A) and the same phase concordance with Wikidata ($K = 0.9695$) in the worked example. Figure 1 shows all

three corpora mapped against the 3° harmonic grid.

The interpretation follows [Silva \(2020\)](#)’s distinction between statistical structure and archaeological explanation. Significance tests evaluate evidence against a null model; archaeological explanation is developed from evidence independent of the orientations themselves, including the wider archaeological record and socio-historical context. The statistical pattern illustrated here is compatible with several mechanisms. These include geographic channeling by trade routes and river systems following the Eurasian corridor, cultural diffusion patterns concentrating stupa construction at historically dominant nodes ([Sen, 2014](#)), and selection effects in the UNESCO inscription process. The hub-centrality result in Section 5 is consistent with this: if major route hubs and hemispherical monuments are both concentrated along the same interior Eurasian corridor, the harmonic proximity of each would be a shared consequence of corridor geography rather than an independent proximity result. Disentangling monument proximity from hub proximity would require a geographic control outside the corridor, which remains a target for future comparative work. Historical interpretations involving deliberate placement would additionally require evidence for pre-modern continental longitude determination at sub-degree precision ([Hunger & Steele, 2019](#); [Neugebauer, 1975](#)). The contribution of the present work is to establish what the directly matched statistical test reports when applied to three differently assembled corpora.

6.3 Limitations

All p -values in Table 2 are conditional on the fixed 3° period and $\sim 35.25^\circ\text{E}$ Levantine corridor phase. Because all three datasets are publicly available open data, it is not possible to determine when or in which corpus the frame was first noticed; we therefore treat all three corpora uniformly as frame-specified rather than attempting to draw a conditional/independent distinction between them. An anchor-sweep correction evaluates how unusual the Levantine corridor is globally. The sweep tests every longitude from 0° to 360°E on a uniform 0.01° grid (36,000 anchors), recording the A+ count ($p_0 = 10\%$ geometric null) for the full UNESCO corpus at each anchor. The observed Levantine peak (132 A+ hits) falls at the 99.7th percentile of the global distribution (top 0.3% of all trial anchors). A max-statistic permutation test (10,000 uniform-null draws) then corrects for having selected the best of 36,000 anchors; the resulting discovery-corrected p -value is 0.0753 (marginal, $z = 1.65$), providing a boundary condition on phase-free discovery claims.

The Bonferroni correction in Null B ($k = 3$, $\alpha_{\text{adj}} = 0.0167$) covers the three window widths ($\pm 2^\circ$, $\pm 5^\circ$, $\pm 10^\circ$) tested in that null. No additional proximity tests were run within Null B beyond these three windows.

7 Conclusion

This paper clarifies the statistic appropriate to a repeated-line longitude hypothesis. Rayleigh concentration and harmonic proximity are related but distinct descriptors of a modulo- T phase distribution: Rayleigh summarizes mean circular concentration, while the geometric-null binomial tests thresholded proximity to any harmonic under the benchmark $p_0 = 2d/T$. The period sweep shows this distinction empirically, with Rayleigh-significant periods lacking proximity enrichment and the enriched 3.0° period lacking

significant Rayleigh concentration. Applied under a fixed operational frame — fixed period, corridor phase, thresholds, and corpora — the binomial framework illustrates a consistent corpus-level proximity pattern across three hemispherical-monument datasets at the 3° period and ~35.25°E Levantine corridor phase. This serves as the worked-example illustration of the method. Following Silva (2020), the worked example is best read as a descriptive spatial exercise: the statistical tests describe the proximity structure of these open-data corpora under the fixed frame, while questions of mechanism, intentionality, and historical transmission require independent archaeological and socio-historical evidence. The paper’s contribution is the matched geometric-null framework, the demonstrated Rayleigh–binomial dissociation, and a reproducible open-data example that future comparative work can evaluate.

All data, code, and audit files are openly available at Zenodo (Tenenbaum 2026; <https://doi.org/10.5281/zenodo.19938283>).

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Data Availability

All data, code, and audit files: Zenodo (Tenenbaum 2026; <https://doi.org/10.5281/zenodo.19938283>). UNESCO XML: UNESCO World Heritage Centre 2024.

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Conflict of Interest

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